



American International University- Bangladesh (AIUB)
Faculty of Engineering (EEE)

Course Name:	Digital Logic Circuits	Course Code:	EEE 3101		
Semester:	Spring 24-25	Assignment Name:	OBE Assignment		
Marks:	30	Term:	Final		
Faculty:	Dr. Shahriyar Masud Rizvi	Section	G		
Student Name:	Basudeb Kundu	Student ID:	23-50856-1	Serial :	
Submission Date:	15 – 06 - 2025	Department:	Bsc CSE		

COs/ CLOs Number	COs/CLOs Statements	K	P	A	Assessed Program Outcome Indicator	BNQF Indicato r	Teaching- Learning Strategy
CO2	Develop a digital system with conflicting requirements of a complex engineering problem.	3	P1,P2, P6	-	P.a.3.C3	FS.1	OBE Assignment (Final Term)

Instructions Related to Use Variables:

Note that this problem uses the variables a, b, c, d, e, f, g and h, which are the digits of your student ID (ab-cdefg-h).

Any value in student ID comes '0' will be replaced by '(a+b+h)'

a	b	-	c	d	e	f	g	-	h
2	3	-	5	0	8	5	6	-	1

Marking Rubrics (to be filled by Faculty):

Complex Problem	Task	Assessment Criteria	Evaluation Criteria				Marks
P1, P2, P6	I.	Outline the necessary steps of the block diagram and flow diagram.	Excellent (2.5)	Good (2-1.5)	Average (1-5)	Poor (0)	
			All the steps have been identified and are in the correct sequence	Not all the steps have been identified and in the correct sequence	Few steps have been identified and in the correct sequence	All the steps have been found in the wrong sequence.	
		Design counter with the necessary diagram	Excellent (2.5)	Good (2-1.5)	Average (1-5)	Poor (0)	
			All the designs were accurate and working	Not all the designs were accurate and working	Few designs were accurate and working	All the designs were wrong and not working	
	II.	Design of System with timer circuit 1 & 2	Excellent (12-10)	Good (9.5-6)	Average (5.5-3)	Poor (2.5-0)	
			Circuit design is correct and complies with the problem	Circuit design is correct, but does not comply with the problem	Circuit design is incorrect and does not comply with the problem	Circuit design is wrong and does not comply with the problem	
	III.	FSM design and HDL Code	Excellent (8-7)	Good (6.5-4)	Average (3.5-2)	Poor (1.5-0)	
			Design is correct and complies with the requirements, with no or minor calculation errors.	Design is correct and complies with the requirements, but with major calculation errors.	Design has major flaws which does not comply with the requirements, but with minor calculation errors.	Design has major flaws which does not comply with the requirements, with major calculation errors.	
	IV.	Limitations of the developed system	Excellent (5-4)	Good (3.5-2)	Average (1.5-1)	Poor (0)	
			Provides limitations and analyzes the performance correctly	Provides limitation only	Provides improper limitations and performance analysis	Provides no limitation and gives no performance analysis	

Marks Obtained:

(i) ID = 23-50856-1

a b - c d e f g - h

a	b	-	c	d	e	f	g	-	h
2	3	-	5	0	8	5	6	-	1

$$\begin{aligned}\text{Max con Capacity} &= a + g + h \\ &= 2 + 6 + 1 = 9 \text{ cony}\end{aligned}$$

Decimal to binary - $(9)_{10} \rightarrow (1001)_2$

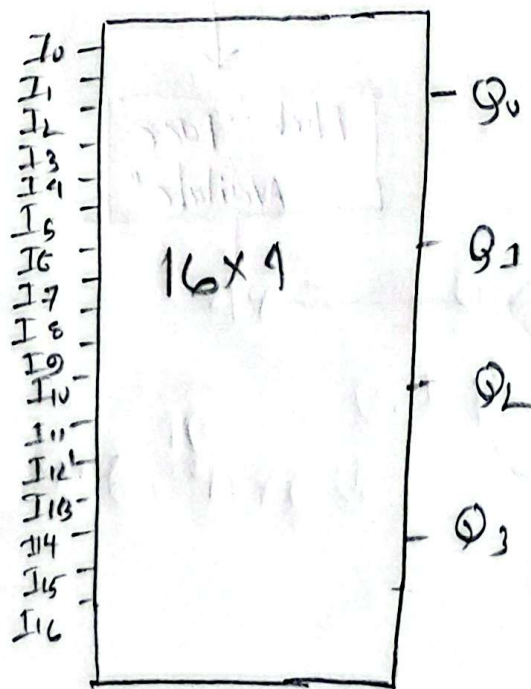
We will create block diagram for 4 bit including comparator

Step-1 : Basudeb Kundu

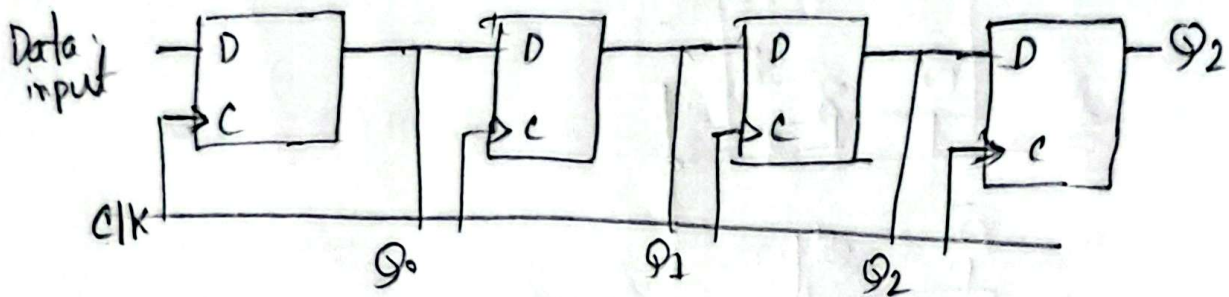
Now, I will take 6 as a input from the keypad.

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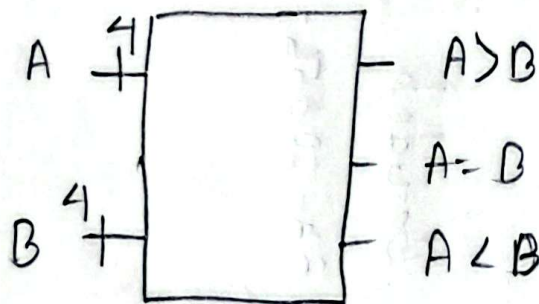
Step-2 : convert this decimal number into Binary using (16×4) encoder.



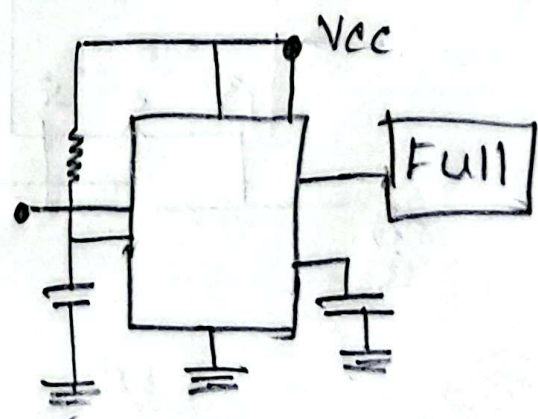
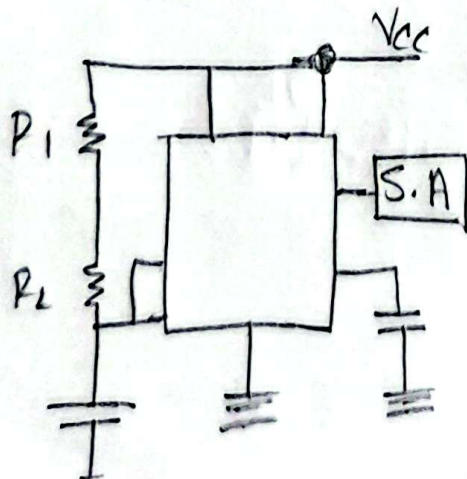
Step-3 : 4 bit register to hold the value read this shift register is serial input and parallel output (SIPO)



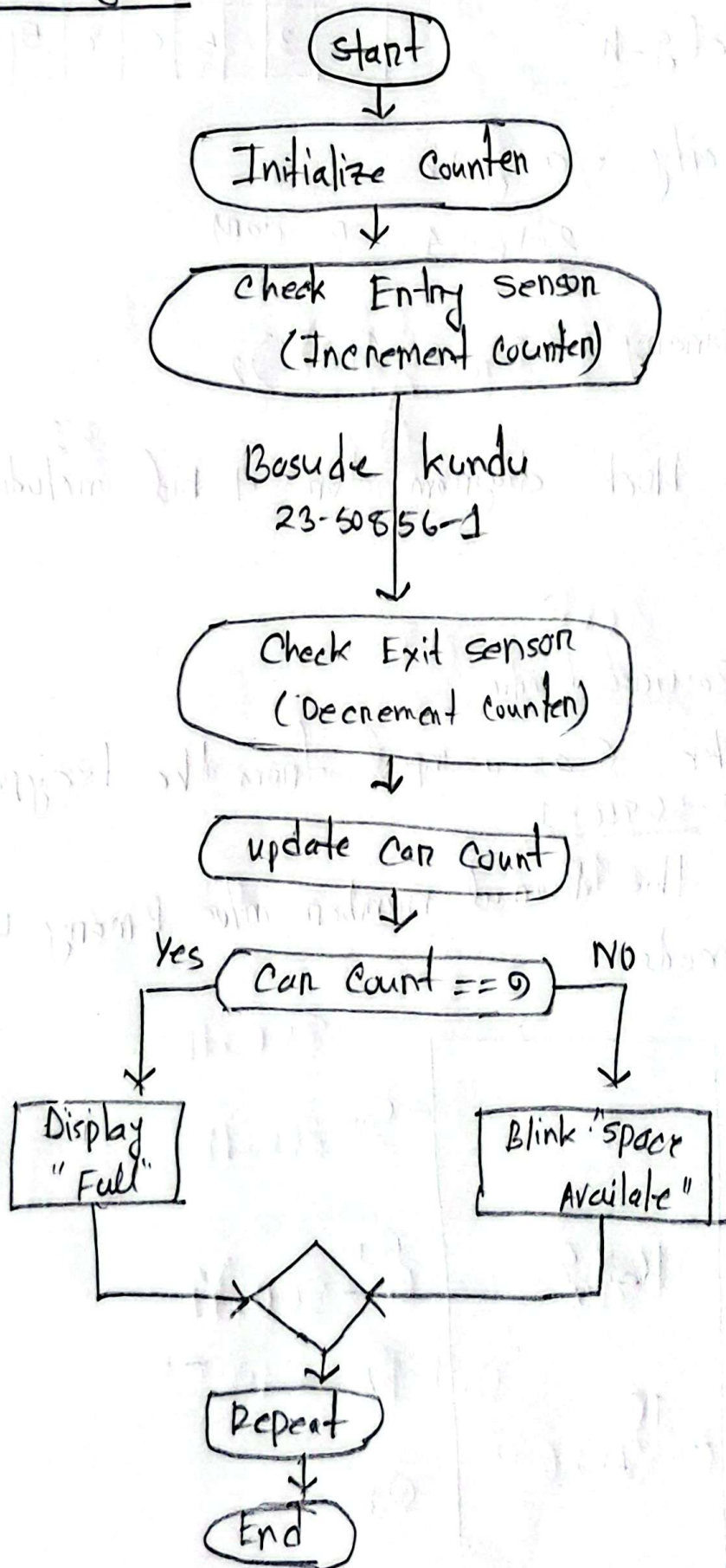
Step-4 4 bit comparator to compare the binary value A and B
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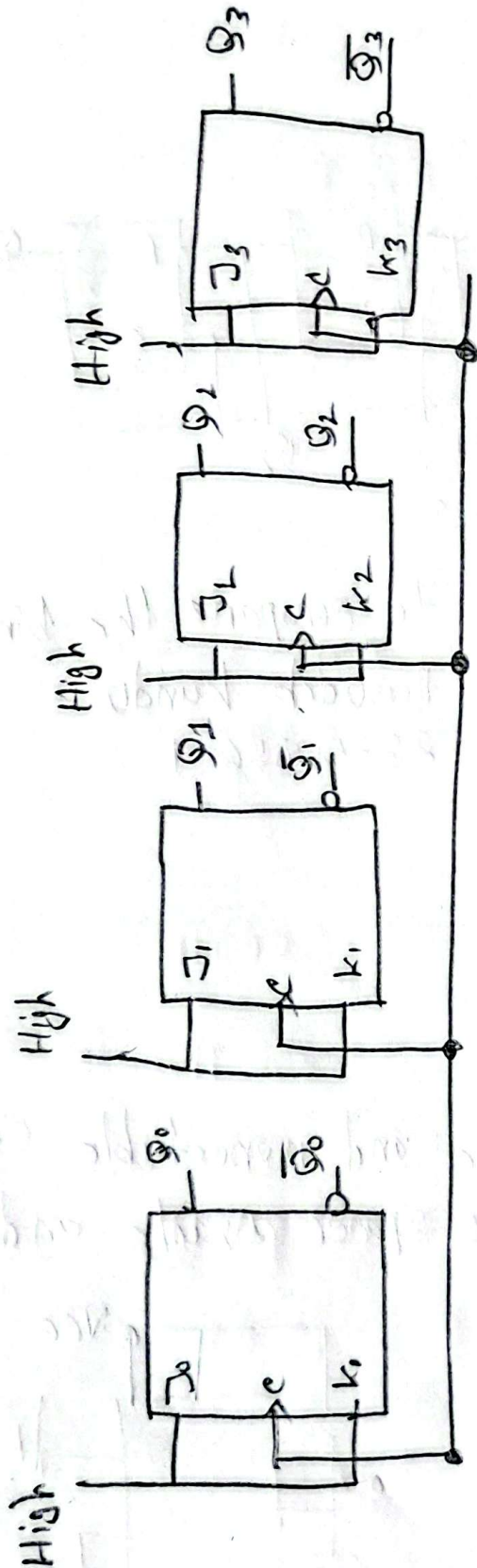
Step-5 : Now use astable and monostable 555 timer circuit to display space available and full.



Flow Diagram



Architecture of 4-bit Synchronous up/down Counter -



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(ii) Calculating the derived values,
For 555 timer, astable,

Given,

$$\text{Duty cycle (D\%)} = 100 - P$$

$$P = (a + c + d + e + f) \times 2$$

$$= (2 + 5 + 0 + 8 + 5) \times 2$$

if any value
"0" then

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$$(a + b + h) = 2 + 3 + 4 = 6$$

$$\begin{aligned} \text{Now, } P &= (a + c + d + e + f) \times 2 \\ &= (2 + 5 + 6 + 8 + 5) \times 2 \\ &= 52\% \end{aligned}$$

$$\therefore D = 100 - P = (100 - 52) = 48\%$$

$$\begin{aligned} \text{Frequency; } f &= (a + d + 2) \\ &= 2 + 6 + 2 = 10 \text{ Hz} \end{aligned}$$

$$T = \frac{1}{f} = \frac{1}{10} = 0.1 \text{ s} = 100 \text{ ms}$$

We know,

$$\text{Duty cycle, } D = \frac{T_H}{T_H + T_L} = \frac{T_H}{T}$$

$$\begin{aligned} \therefore T_H &= D \times T = \frac{48}{100} \times 100 \text{ ms} \\ &= 48 \text{ ms} \end{aligned}$$

The display will be "ON" for $T_H = 48 \text{ ms}$

Again, $T = T_H + T_L$

$$\Rightarrow T_L = T - T_H = 100 \text{ ms} - 48 \text{ ms} \\ = 52 \text{ ms}$$

The display will be "OFF" for $T_L = 52 \text{ ms}$

For : 250 μF capacitor

$$C = 250 \mu\text{F}$$

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$$T_H = 0.693 \times (R_1 + R_2) \times C$$

$$T_L = 0.693 \times R_2 \times C$$

For : $T_L = 51.99 \text{ ms}$

$$R_2 = \frac{T_L}{0.693 \times C} = \frac{52}{0.693 \times 250 \times 10^{-3}}$$

$$= 300.12 \Omega$$

$$R_1 = \frac{T_H}{0.693 \times 250 \times 10^{-3}} - 300.12$$

$$= \frac{48}{0.693 \times 250 \times 10^{-3}} - 300.12$$

$$= 9.65 \Omega$$

For

470 μ F capacitor

$$R_2 = \frac{51.00 \times 10^{-3}}{0.693 \times 470 \times 10^{-6}} = 150.56 \Omega$$

$$R_1 = \frac{48 \times 10^{-3}}{0.693 \times 470 \times 10^{-6}} - 150.56$$

$$= -12.18 \cong 12.18$$

For, 50 μ F

$$R_2 = \frac{51.00 \times 10^{-3}}{0.693 \times 50 \times 10^{-6}} = 1500.43$$

$$R_1 = \frac{48 \times 10^{-3}}{0.693 \times 50 \times 10^{-6}} = 1385.28$$

The capacitor of 250 μ F is the best choice from the given capacitor (250 μ F, 470 μ F, 50 μ F).

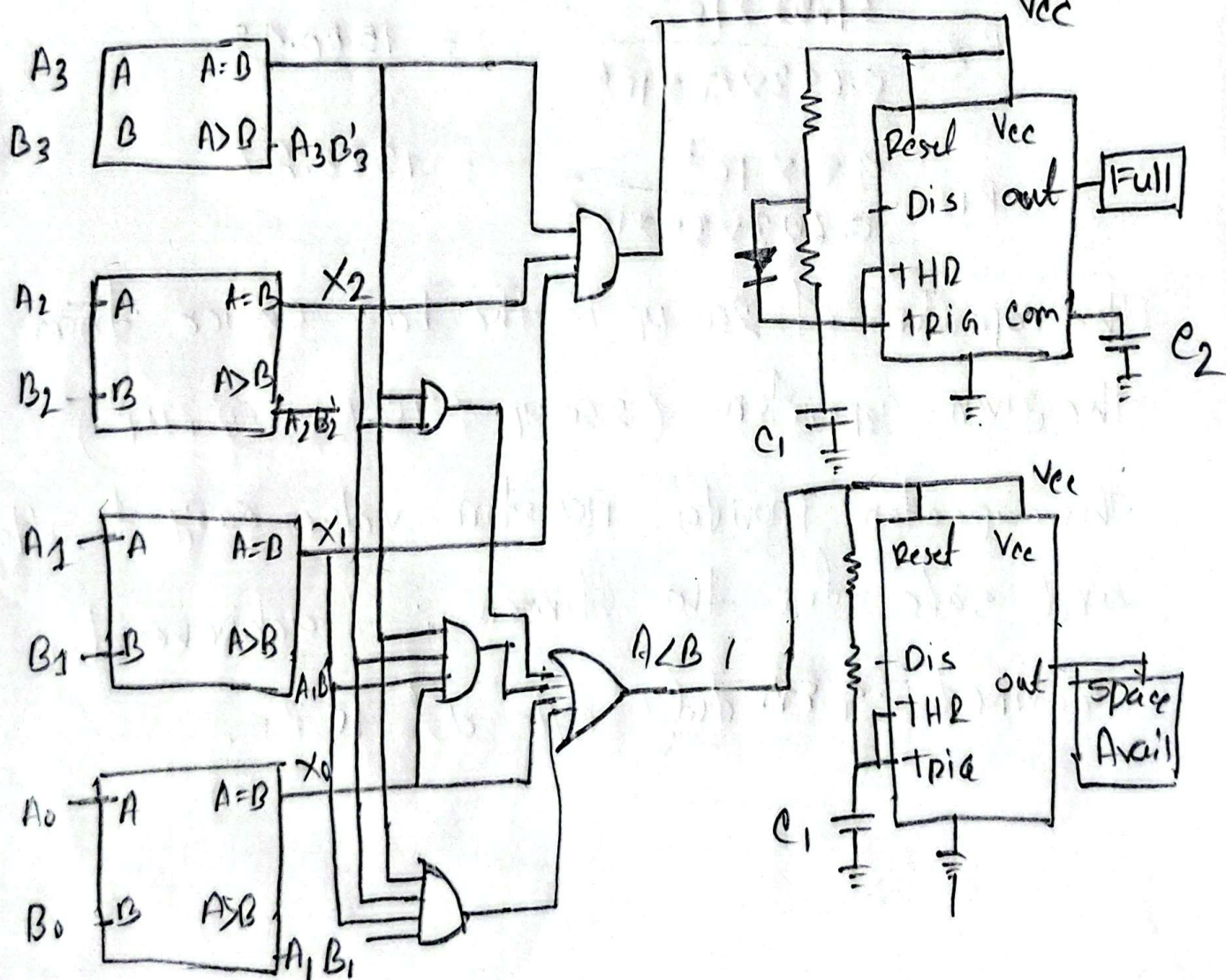
This capacitor provides resistor value easy to obtain and safe for the timer. It achieves the required 48V duty cycle at 10 Hz.

With an average timing of 1 ms "ON" and
ms OFF. The MF for low resistors
drawing too much current. MF needs very high
resistor.

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$$A > B = A_3 B_2' + X_3 A_2 B_2' + X_3 X_2 A_1 B_1' + X_3 X_2 X_1 A_0 B_0$$



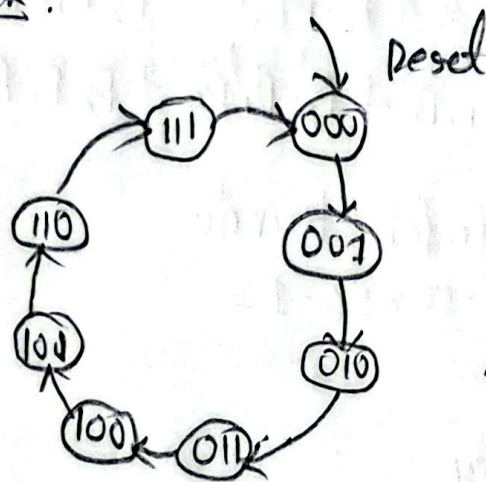
(iii) As $T_H = 46 > 4$

$T_H = 4$

Again $T_L = 52 > 4$

$T_L = 4$

State Diagram:



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State table

Reset	P state			N state			Output	
	P_{S2}	P_{S1}	P_{S0}	N_{S2}	N_{S1}	N_{S0}	A	B
1	X	X	X	0	0	0	0	0
0	0	0	0	0	0	1	1	0
0	0	0	1	0	1	0	1	0
0	0	1	0	0	1	1	1	0
0	0	1	1	1	0	0	1	0
0	1	0	0	1	0	1	0	1
0	1	0	1	1	1	0	0	1
0	1	1	0	1	1	1	0	1
0	1	1	1	0	0	0	0	1

Equation :-

$$N_{s0} = \overline{\text{Reset}} (\bar{P}_2 \cdot \bar{P}_1 \cdot \bar{P}_0 + \bar{P}_2 \cdot P_1 \cdot \bar{P}_0 + P_2 \cdot \bar{P}_1 \cdot \bar{P}_0 + P_2 \cdot P_1 \cdot \bar{P}_0)$$

$$N_{s1} = \overline{\text{Reset}} (\bar{P}_2 \cdot \bar{P}_1 \cdot P_0 + \bar{P}_2 \cdot P_1 \cdot P_0 + P_2 \cdot \bar{P}_1 \cdot P_0 + P_2 \cdot P_1 \cdot P_0)$$

$$N_{s2} = \overline{\text{Reset}} (\bar{P}_2 \cdot P_1 \cdot \bar{P}_0 + \bar{P}_2 \cdot P_1 \cdot P_0 + P_2 \cdot \bar{P}_1 \cdot \bar{P}_0 + P_2 \cdot P_1 \cdot \bar{P}_0)$$

$$A = (\bar{P}_2 \cdot \bar{P}_1 \cdot \bar{P}_0 + \bar{P}_2 \cdot \bar{P}_1 \cdot P_0 + \bar{P}_2 \cdot P_1 \cdot \bar{P}_0 + \bar{P}_2 \cdot P_1 \cdot P_0)$$

$$B = (P_2 \cdot \bar{P}_1 \cdot \bar{P}_0 + P_2 \cdot \bar{P}_1 \cdot P_0 + P_2 \cdot P_1 \cdot \bar{P}_0 + P_2 \cdot P_1 \cdot P_0)$$

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Venilog HDL Code :-

module TLC-BINARY

(input wire clk, rst, output wire A, B);

// Internal signal definitions.

reg [2:0] P-STATE

wire [2:0] N-STATE

// Next state logic

assign N-STATE[0] = ~rst & ((~P-STATE[2]

& P-STATE[1] & P-STATE[0]))

| (~P-STATE[2] & P-STATE[1] &

~P-STATE[0]) | (P-STATE[2] &

~P-STATE[1] & ~P-STATE[0]) | (

P-STATE[2] & P-STATE[1] & P-STATE[0])

assign N-STATE[1] = 1 PSTATE[1] = PSTATE[2]

$$\begin{aligned} & P\text{-STATE}[1] \wedge P\text{-STATE}[0]) \wedge \\ & (\neg P\text{-STATE}[2] \wedge P\text{-STATE}[1] \wedge \neg P\text{-STATE}[0]) \\ & \wedge (P\text{-STATE}[2] \wedge \neg P\text{-STATE}[1] \wedge \\ & P\text{-STATE}[0]) \wedge (P\text{-STATE}[2] \wedge P\text{-STATE} \\ & [1] \wedge \neg P\text{-STATE}[0]))); \end{aligned}$$

```

again N-STATE[0] = ~ RSTATE[0] & (MP-STATE[2] &
P-STATE[1] & RSTATE[0]) | (P-STATE[2]
& ~ P-STATE[1] & ~ R-STATE[0]) | (P-
STATE[2] & ~ P-STATE[1] & P-STATE[0]) |
(P-STATE[2] & P-STATE[1] & R-STATE
[0]);

```

// Present state Register
always @ (posedge clk)

begin

$P\text{-STATE}[v] \leq N\text{-STATE}[v];$

P-STATE [1] < N-STATE [1];

$$P\text{-STATE}[2] < N\text{-STATE}[2]$$

end

//output logic

assign A = ((~P_STATE[2] & P_STATE[1] &
 ~P_STATE[0]) | (~P_STATE[2] & ~
 P_STATE[1] & P_STATE[0]) | (~P_STATE[2]
 & P_STATE[1] & ~P_STATE[0]) | ~P_STATE & P
 - STATE[1], & P_STATE[0]))

assign B = ((P_STATE[2] & ~P_STATE[1] &
 ~P_STATE[0]) | (P_STATE[2] & ~P-
 STATE[1] & P_STATE[0]) | P_STATE
 [2] & P_STATE[1] & ~P_STATE[0])
 | (P_STATE[2] & P_STATE[1] &
 P_STATE[0]));

Endmodule -

(iv) Limitation of the current Design :-

1. No Power Backup: if the system restart (due to the power failure) the car count is lost

2. Sensor Bounce and false triggering

Entry / Exit : Sensors may generate multiple transitions. This can count.

3. Limited capacity (3 bit): A 3 bit register can store maximum of 8 cars (from my id) Not suitable for large parking manager.

4. 555 timer limitations: Astable 555 can't be directly enable / disabled via logic input.

monostable 555 can't hold indefinitely needs continuous triggering or logic override.

5. No Display lock mechanism: The segment display may glitch during signal transition without latching.

possible solution to the Limitations at the¹⁴
current Design: -

- (i) Use non-volatile memory to store the car count
- (ii) Use a battery-backed register on microcontroller
- (iii) Use up-to-isolated sensor for higher reliability
- (iv) Expand the counter to 8 bits (0-255) cars.
- (v) Use a BCD counter with Display decoder for direct output to 7 segment display.
- (vi) Add a logic AND gate to enable/disable the timer output based on comparison logic.
- (vii) Use a D-type latch between the timer output and the 7 segment display.

Higher number at car 235 (8 bit)

Then change:

- (i) Encoder (8 bit)
- (ii) Counter (8 bit)
- (iii) Comparator (8 bit)

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